

AMENDMENT NO. 1 DECEMBER 2010
TO
IS 13805 : 2004 GENERAL STANDARD FOR QUALIFICATION AND
CERTIFICATION OF
NON-DESTRUCTIVE TESTING PERSONNEL — SPECIFICATION

(First Revision)

(Page 13, clause 8.4) — Substitute the following for the existing:

‘8.4 Renewal

8.4.1 After the period of validity, certification may be renewed by the National Certifying Body, directly or through an authorized qualifying body, for an additional period of five years from the date of expiry provided the individual meets the following criteria:

- a) Provides evidence of satisfactory visual examination, and
- b) Provides evidence of continued satisfactory work activity without significant interruption.

Applications for renewal can be made up to 6 months before the date of expiry. Up to 6 months after the certificate expiry date, a candidate’s certification status can be reactivated.’

(Page 13, clause 8.4.2) — Delete.

(Page 13, clause 8.5) — Renumber the clause as **8.6**.

(Page 14, clause 8.6) — Renumber the clause as **8.5** and substitute the following for the existing:

8.5 Recertification

8.5.1 General

In case the candidate fails to renew his existing certificates, to regain certification for that method (and sector), the candidate shall be required to apply for fresh certification and has to undergo the appropriate method of examination (general and specific) including practicals in the case of Level I and II.

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Prior to completion of each second period of validity, or at least every ten years, the certified individual may be recertified by the certification body for a similar period, provided the individual meets the criterion in **8.4** for renewal and meets the following conditions, as applicable.

8.5.2 Level I and II

The individual shall successfully complete a practical examination that assesses ongoing competence to carry out within the scope of the certificate, according to the following:

- a) See Table 8 for guidance on the subjects to be covered and their percentile weighting in the practical examination. If the individual fails to achieve a grade of at least 70 percent for each specimen tested, two retests of the recertification examination shall be allowed within 12 months of the first attempt at the recertification examination; and
- b) In the event of failure in the two allowable retests, the individual shall not be recertified and, in order to regain certification for that level, sector and methods, the individual shall apply for new certification. If the individual holds valid certification in a different sector of the same method, an exemption in the general examination will be allowed.

8.5.3 Level III

In the case of Level III personnel recertification by Indian Society for Non-destructive Testing is intended to apply only to individuals who maintain continued active employment in NDT Level III functions and demonstrate efforts to keep abreast of the technology in the method(s) for which recertification is sought.

Recertification can be obtained by application subject to the following conditions:

- a) Reaffirmation of the 'Code of Ethics' for Level III Non-destructive Testing Personnel certified by ISNT and as prescribed in the NCB Quality Manual;
- b) Continued active employment in Level III functions as related to the NDT Method(s) for which recertification is sought;

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- c) Such employment must have covered at least 36 months during the valid certification period but not necessarily 36 consecutive months;
- d) At least 12 of the 24 months immediately preceding the expiration of the certification must have been spent in Level III functions; and
- e) If work experience during the certification period did not include all methods for which recertification is sought, at least two additional points shall be obtained in each method for which there was no work experience.

The demonstration of an effort to keep up-to-date, contribute to knowledge or maintain continued growth in the method(s) for which the individual is certified or to expand knowledge in related technologies by obtaining a minimum number of points during each 5 year period of certification as given in Table 11.

To be recertified, it is necessary to gain a minimum of 30 points in a 5 year period. A maximum of 10 points per year will be allowed.

(Page 14, Table 11) — Add reference to clause **8.5.3**.

(Page 15, Annex A) — Add the following matter after the table:

Basic Examination Level III

This Basic Examination will cover three (3) main areas:

1.1 Personal Qualification and Certification

The NDT Personnel should have knowledge of certification body's qualification and certification system based on this Standard (IS 13805).

1.2 General Knowledge of Materials, Fabrication Processes and Product Technology

1.3 Fundamentals of Material Technology

- a) Structure of metals and alloys;
- b) Physical and mechanical properties of metallic materials; and

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- c) Behaviour of materials in service, Modes of failures:
 - i) Corrosion,
 - ii) Fatigue,
 - iii) Wear,
 - iv) Overload, and
 - v) Brittle fracture.

1.4 Fundamentals of Fabrication and Product Technology

- a) Metal processing — Castings, Mechanical working (Forging, Rolling, Extrusion, Drawing), Heat Treatment, Welding, Powder Metallurgy;
- b) Non-metals and composite materials processing; and
- c) Origin & Nature of Defects associated with each processing method.

1.5 Dimensional Metrology

Fundamental Units & Standards — Gauging — Interferometry

2.0 General Familiarity with the Following NDT Methods

<i>NDT Methods</i>	<i>Abbreviations</i>
Acoustic Emission Testing	AT
Electromagnetic Testing	ET
Penetrant Testing	PT
Leak Testing	LT
Magnetic Particle Testing	MT
Neutron Radiographic Testing	NRT
Radiographic Testing	RT
Infrared/Thermal Testing	IRT
Ultrasonic Testing	UT
Vibration Analysis Testing	VAT
Visual Testing	VT

Fundamentals – Principles/Theory – Sensors/Sources – Equipment and Material – Techniques – Detection – Electronics/Instrumentation – Interpretation & Evaluation of Test Results – Codes and Standards – Safety – Documentation – Application

This is applicable to all the mentioned NDT methods.

(MTD 21)

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